PROJECT TITLE: HUMAN RESOURCE, PAYROLL AUTOMATION AND JOB PROCESSING SYSTEM

1. ABSTRACT

In many organizations, HR operations, payroll calculations, and project tracking are handled separately using paper registers, messenger, and Excel sheets. This manual workflow wastes time, causes mistakes in salary calculations, misses unpaid leaves, and makes it hard for managers to know the real daily progress of ongoing company projects. In this project, I have developed an Automated Human Resource, Payroll, and Project Tracking Management System using Python Django, MySQL, and a Bootstrap frontend. The system centralizes employee records, tracks daily attendance, manages leaves, automates monthly salary disbursement, and includes an active Project Progress Tracking module.

The system provides 3 role-based portals: Admin, HR Manager, and Employee. Admin creates accounts for multiple HR managers. HR onboards new employees, uploads documents, creates company projects, assigns them to employees, and monitors project completion in real time. Employees log in to update their project progress percentage, submit work remarks, view their profile, track leave balances, apply for leaves, and clock in/out with Office Wi-Fi IP verification to prevent fake attendance. The system also features a Payroll Automation Engine: with one click, HR calculates monthly salaries with automatic Loss of Pay (LOP) for absences, deducts PF and taxes, and generates official downloadable PDF pay slips. This makes company operations organized, productive, and paperless.

2. OBJECTIVES:

1. To store employee records, departments, designations, bank details, and uploaded documents in a single MySQL database.
2. To enable online clock-in and clock-out with automatic office wi-fi IP validation so employees cannot mark attendance from home.
3. To let employees apply for leaves online, automatically track remaining leave balances, and allow HR to approve or reject them.
4. To automatically calculate gross pay, Loss of Pay (LOP) for absent days, provident fund and taxes without manual Excel math.
5. To allow HR to process company-wide monthly payroll.
6. To automatically generate official, downloadable PDF salary slips.
7. To separate permissions between Admin, HR, and Employees so sensitive financial data remains confidential.
8. To enable HR to create and assign projects to employees, and allow employees to update daily progress percentages and completion remarks.

3. EXISTING SYSTEM:

Currently, companies manage HR, payroll, and project updates using disconnected tools: paper attendance books, Excel files, and verbal or WhatsApp status updates.

Limitations:

1. No Clear Project Visibility: Managers must constantly call or message employees on WhatsApp to ask for project updates, leading to lost details and missed deadlines.
2. Human Math Mistakes: Manual calculation of basic pay, allowances, overtime, and deductions in Excel frequently causes salary errors.
3. Time-Consuming Month-End Work: HR spends days manually cross-referencing attendance registers with Excel salary sheets.
4. Missed Unpaid Leaves: If an employee takes unapproved leave and HR forgets to note it down, the employee gets paid for days they didn't work.
5. Proxy Attendance Risk: Traditional registers and basic web forms allow employees to mark proxy attendance or clock in from home.
6. No Self-Service Portal: Employees have no single dashboard to view their salary slips, apply for leaves, or report daily work progress

4. PROPOSED SYSTEM

The proposed system is a secure, web-based platform that replaces spreadsheets with database-driven automation.

How System Works:

1. Attendance with Wi-Fi IP Check: When an employee clicks "Clock In", the system checks their IP address. It only marks them "Present" if connected to the Office Wi-Fi; otherwise, it blocks the attempt.
2. Leave Management & Auto-Deductions: Employees apply for leaves online. When HR approves a leave, the employee's remaining leave balance updates automatically.
3. HR creates projects, sets deadlines, and assigns team members and employees log in to view their assigned projects, update completion percentages (e.g., 25%, 50%, 100%), and write daily status notes.
4. Automated Loss of Pay (LOP) system calculates daily basic pay: Daily Pay = Basic Salary/Days in Month, it multiplies this by unapproved absent days and deducts it from the gross pay automatically.
5. Statutory Deductions: Automatically applies 12% Provident Fund (PF) and income tax rules.
6. One-Click Monthly Payroll: HR selects the month (e.g. 2026-10) and clicks "Process Payroll". The engine computes all employee salaries and stores them in MySQL within seconds.
7. Instant PDF Pay slips: Both HR and employees can click "Download PDF" to get a formatted, official salary slip anytime.

5. ROLES AND RESPONSIBILITIES

1. Admin:

• Sets up the core system, departments, and designations.
• Creates user accounts for multiple HR managers.
• Has full access to the Django Admin panel (/admin) to inspect database tables and reset passwords.

2. HR Manager

• Onboards new employees, uploads profile photos and documents and generates login credentials.
• Views and edits employee records and company departments.
• Creates projects, assigns them to employees, and monitors progress bars and status updates.
• Monitors daily attendance and manually overrides absentees if needed.
• Reviews leave requests and approves or rejects them.
• Executes the payroll engine and reviews company-wide payout summaries.
• Downloads individual or batch PDF pay slips.

3. Employee:

• Logs in using their credentials and can change their initial password.
• Views their personal profile, department, and salary breakdown.
• Clocks in and clocks out daily.
• Views remaining leave balances.
• Submits leave requests and tracks approval status in real time.
• Downloads monthly PDF pay slips without needing to contact HR.

6. TECH STACK

Components	**Technology**
Back-end	Python Django
Database	MySQL
Database connector	mysqlclient
Front-end	HTML, CSS, JavaScript
Libraries	reportlab
Security	bcrypt